

2 TREES / THE BOOK

2 trees: Build your (D)Home

Working manuscript

Codex publication sample / current manuscript

A reading edition of the current editable manuscript. Illustrations retain their captions and source notes.

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PART / story / draft

Part I — The idea in the wood

Before a cut list, there was a question: what if the useful shape was already inside the tree?

We begin with the earlier experiments, then arrive at the wedge. This part explains the discovery without pretending that every later detail was obvious at the beginning.

10. Eight sections, sixty-four blanks, sixty accepted members

Quantity	One tree	Two trees
Planned six-foot sections	8	16
Potential blanks at eight per section	64	128
Target allocation	60	120
Potential spare or rejected blanks	4	8

The eight potential spares represent 6.25 percent of the gross 128-piece inventory. If eight blanks are rejected, the remaining 120 exactly meet the count and leave no accepted replacements. At a ten-percent rejection rate, the arithmetic no longer supports a complete 120-member frame from this inventory. Report that shortage while there is still time to change the plan.

Eight six-foot sections contain forty-eight feet of net section length. They cannot all be cut from exactly forty-eight feet while also losing material to separating cuts. If seven interior bucking cuts remove a quarter inch each, the minimum span before outer-end trimming is 48 feet plus 1.75 inches, or about 48.146 feet. Additional cuts at the ends and defect removals require more length. This example declares its kerf assumption; it is not a measurement of the author's saw.

Separate bucking kerf from ripping kerf. The first reduces the longitudinal inventory as sections are separated. The second removes wood from their cross-sections as wedges are produced. Both matter, but subtracting one from the other would mix different dimensions.

One organizational option is to reserve one tree's qualified blanks for A members and the other tree's for B members. That makes the title easy to explain, but assignment should follow the actual inventory. A short or defective A candidate may be perfectly useful in a B position if its material properties and finished dimensions meet the design. Keep its original tree and section ID when reassigning it.

The project's earlier timber demonstration also explores twelve-foot sections split and then divided into shorter members. That is a different cut plan. Its yield cannot be copied onto this eight-by-six-foot plan without accounting for usable trunk length and every additional crosscut.



Figure 1

Eight sections per tree, eight potential blanks per section: 128 gross blanks and a target of 120 members.

Source: Rendered from existing DomeSim model geometry; see two_trees_codex/render.py. Print colors are illustrative.

17. The triangle corner is a physical object

The ideal triangle has vertices where lines meet. Real members have width, depth, and ends that must occupy different space. The project's pinwheel arrangement gives each member a receiving role at one end and a terminating role at the other. One member ends against the side of its neighbor while receiving another member in turn.

Earlier explanatory code describes simplified square-ended pinwheel members. The current raw-wedge simulator carries more detailed corner geometry: the terminating end is solved against its receiving surface, and the other end is clipped to the triangle's envelope. The corresponding cut planes and stock extents should govern the fabrication discussion for this configuration.

This correction matters to the story. Preserving the longitudinal forty-five-degree sector does not imply that every end is a square crosscut. The long radial faces, the panel-to-panel fold, and the member's end cuts are separate geometric questions. A process can preserve the raw sector while still requiring an oblique end cut.

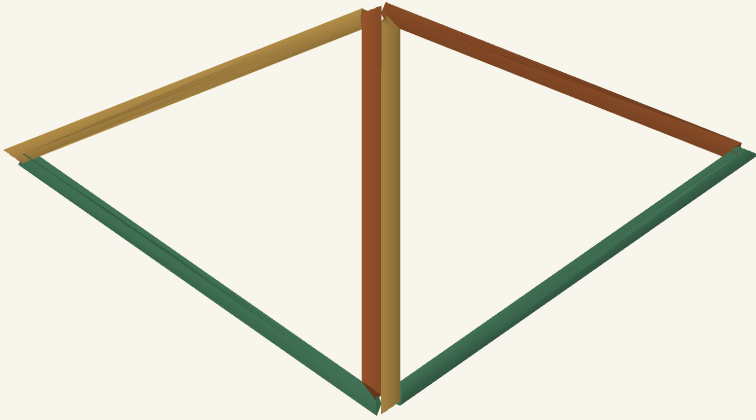
Start with a full-size representative corner using the intended cross-section and orientation. Mark the mathematical reference points and the actual cut surfaces separately. Check the receiving relationship, the available bearing region, the fastening space, and whether either member extends beyond the intended panel envelope.

Move from that corner to one A-A-A panel and one B-A-B panel. Check their reference dimensions and their physical boundaries. A fixture that reproduces one family does not automatically reproduce the other. Before cutting the batch, retain a labeled master example and the configuration that produced it.

2 TREES / PROJECT PLATES

The pinwheel / two neighboring panels

Each butt meets the side of the next member. Each triangle owns three members; a shared axis carries two.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 1

Two separate cyclic pinwheel triangles, each with three members. Two members remain at their shared axis. Colors identify member ownership.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.